WEEK 4 – TUTORIAL

Objectives:

The aim of this tutorial is to engage students in exploring and discussing practical challenges in **scope and schedule management**. Through group-based activities, students will identify risks of scope creep, debate the usefulness of the Work Breakdown Structure (WBS), and analyze scheduling dependencies using real-world project scenarios. The focus is on **critical discussion**, collaboration, and applying project management concepts in context.

Tutorial Activities

- Scope Creep & Risk Analysis
- Work Breakdown Structure (WBS)
- Scheduling Dependencies and Critical Path

Learning outcomes

By the end of this tutorial, students will be able to:

- **LO1.** Understand each of the core project management knowledge areas
- **LO2.** Apply project management knowledge, tools and techniques
- **LO4.** Develop basic proficiency in project management tools and practices

PART A – Scope Creep & Risk Analysis

Three Risks from Scope Creep:

1. **Delays in Delivery** – Additional features extend development time, pushing project beyond original deadline.
2. **Budget Overruns** – Unplanned features require more resources and testing, exceeding original budget.
3. **Reduced Quality** – Adding scope without adjusting resources increases workload, potentially leading to rushed or poorly tested deliverables.

Two Strategies to Manage/Prevent Scope Creep:

- **Formal Change Control Process** – Any new requests go through structured evaluation for impact on time, cost, and resources.
- **Clear Scope Statement & Validation** – Establish inclusions/exclusions early and validate scope at key milestones with stakeholder sign-off.

PART B – Work Breakdown Structure (WBS)

Example Simplified WBS (3 Levels):

1. **Requirements**
    - 1.1 Requirement Gathering
    - 1.2 Requirement Validation
2. **Design**

  - 2.1 System Architecture Design
  - 2.2 UI/UX Design
3. **Development**
   - 3.1 Front-end Development
   - 3.2 Back-end Development
   - 3.3 Integration
4. **Testing**
   - 4.1 Unit Testing
   - 4.2 System Testing
   - 4.3 User Acceptance Testing
5. **Deployment**
   - 5.1 Training & Documentation
   - 5.2 Production Deployment

PART C – Scheduling Dependencies

Activities:

- Requirements (2 weeks) → Design (3 weeks) → Development (6 weeks) → Testing (4 weeks) → Deployment (2 weeks)

AON Diagram:

A → B → C → D → E

Critical Path & Duration:

- Path: A–B–C–D–E
- Total Duration: **17 weeks** (2+3+6+4+2)

If Development = 8 weeks:

- New Duration = 19 weeks
- Critical Path remains A–B–C–D–E